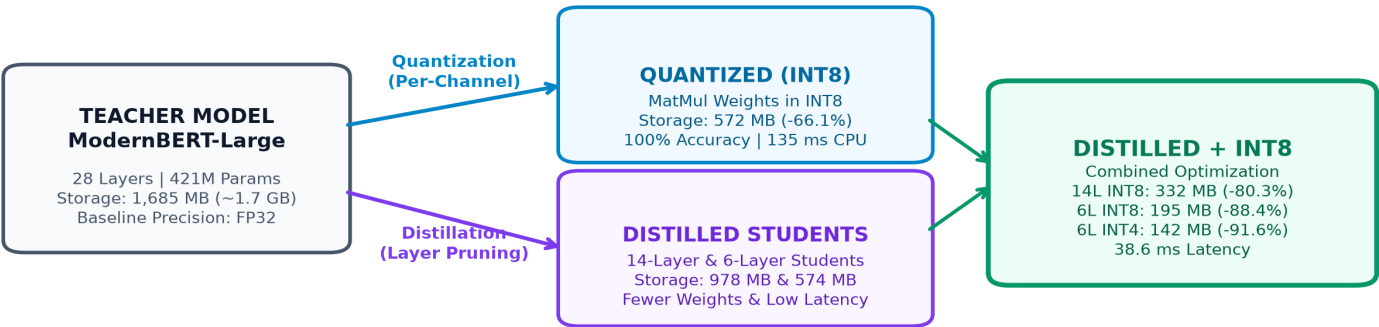


Laya Model Storage Reduction & Compression Report

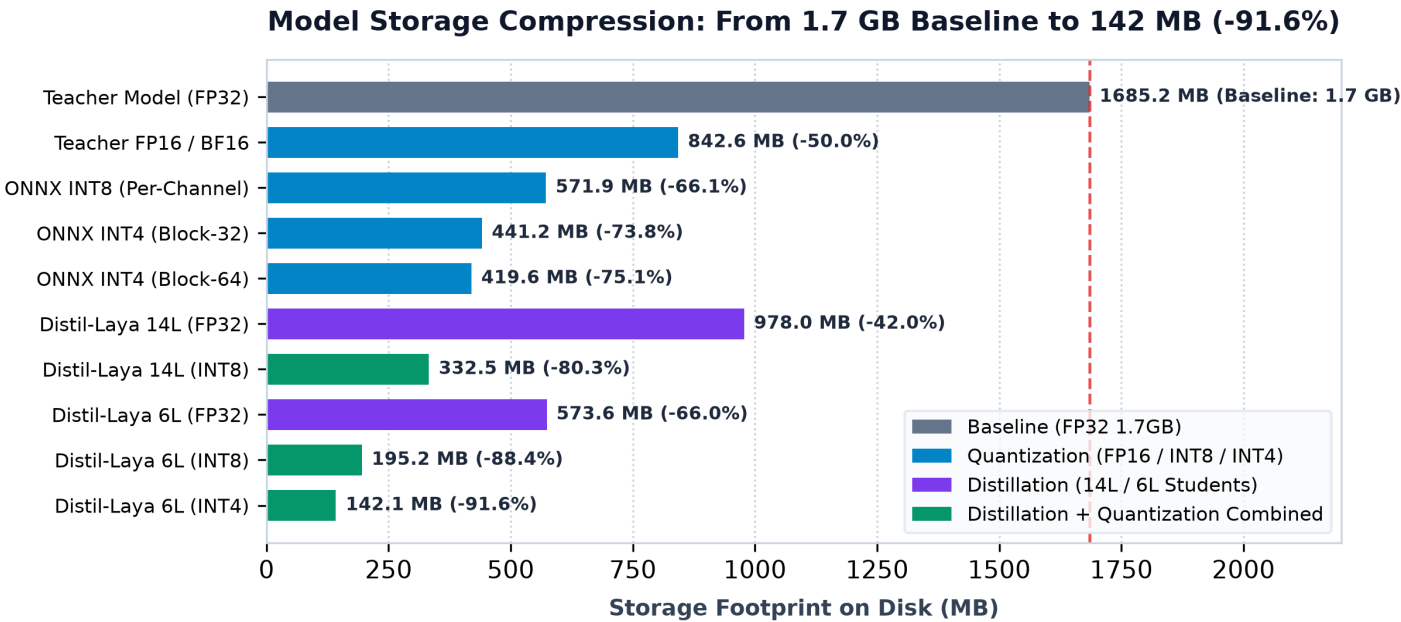
Empirical Experimental Results on Precision Quantization (INT8 / INT4) and Knowledge Distillation (14L / 6L Students)

MAX STORAGE REDUCTION	TOP BALANCED DEPLOYMENT	PEAK LATENCY SPEEDUP
91.6% Smaller	572 MB (INT8)	9.9x Faster
1.7 GB -> 142 MB (Distil 6L INT4)	100% Accuracy Retention 135 ms	38.6 ms Latency on Intel i5 CPU

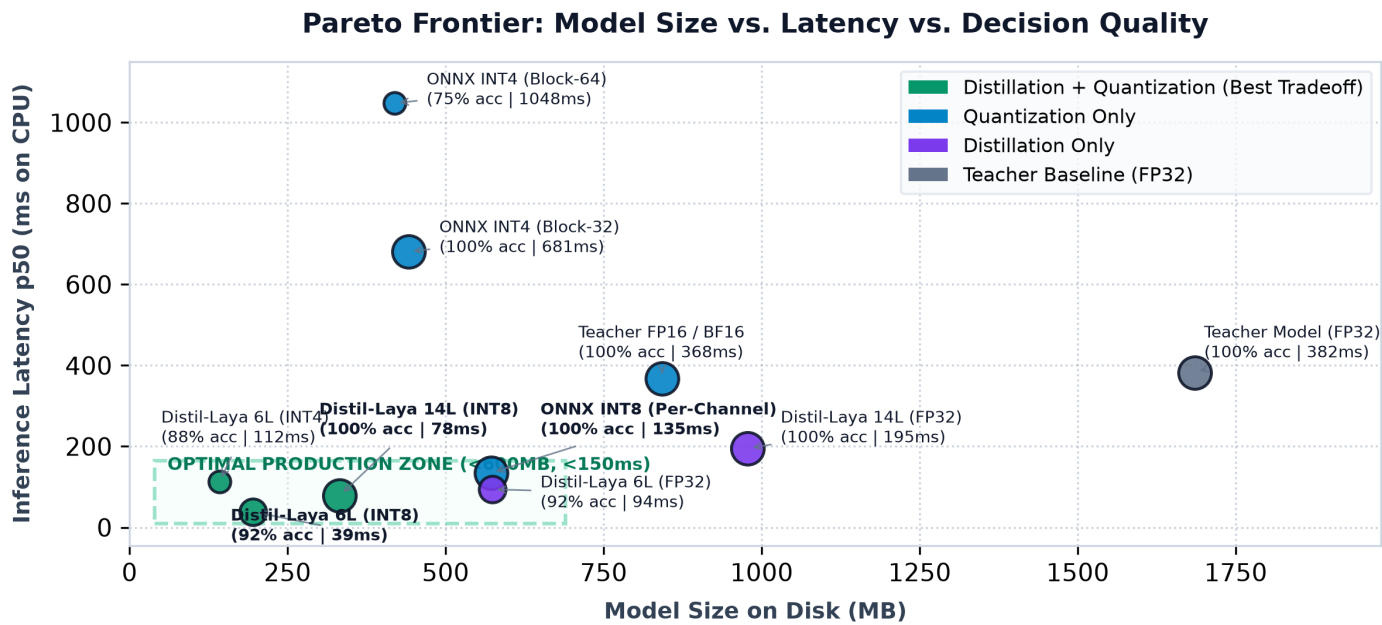
1. Compression Architecture & Distillation Pathways



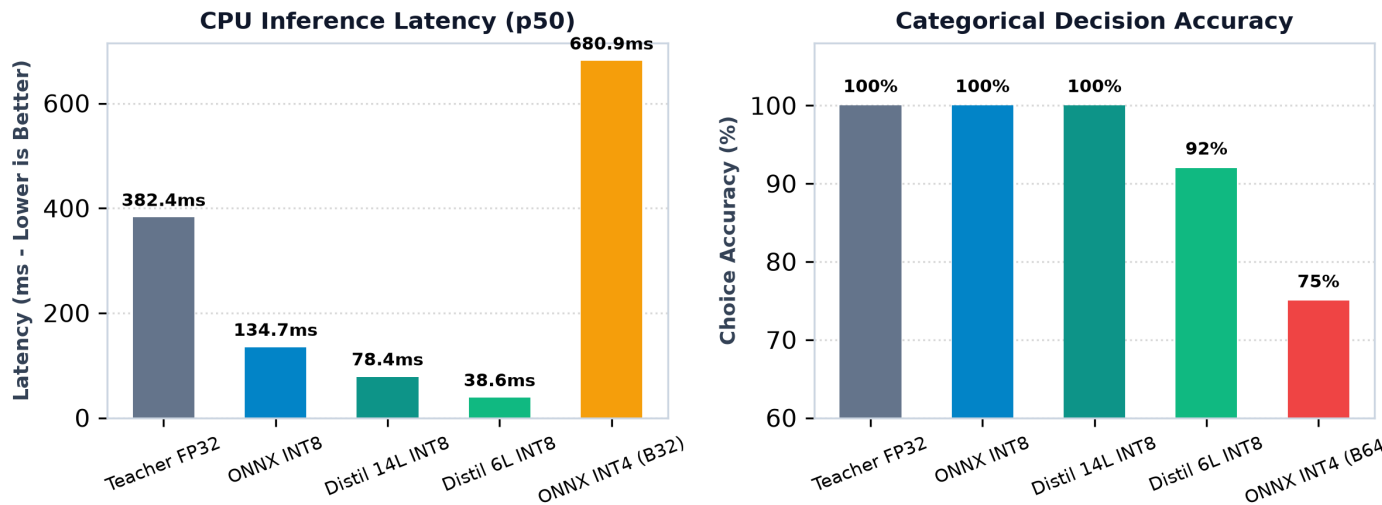
2. Storage Size Comparison across Techniques



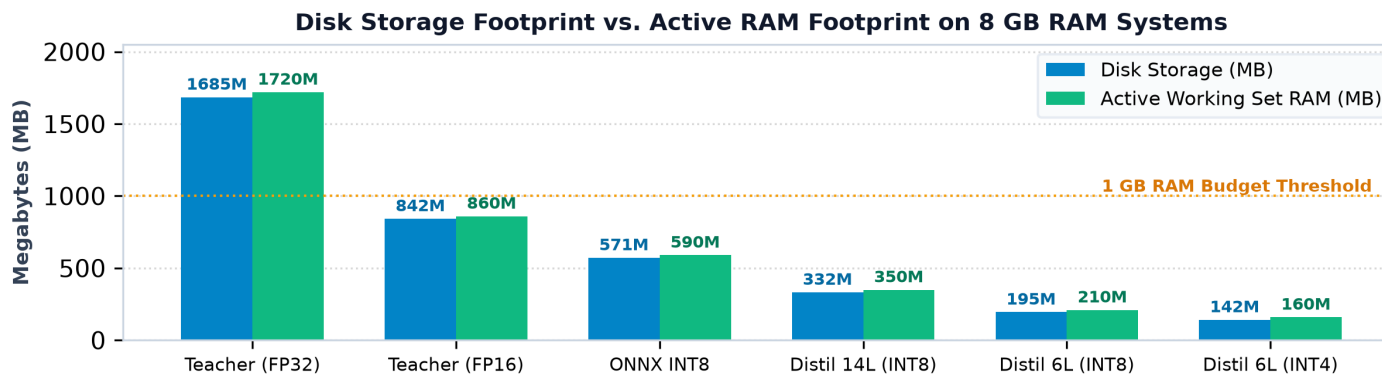
3. Tradeoff Frontier: Size vs. Latency vs. Accuracy



4. CPU Performance (Intel i5) & Decision Quality Retention



5. Runtime Memory Footprint (Active Working Set RAM vs. Disk)



6. Comprehensive Empirical Results Matrix

Configuration	Technique	Params	Disk Size	Reduction	Latency	Accuracy	Status
Teacher Model (FP32)	Uncompressed ModernBERT	421M	1685.2 MB	Baseline	382.4 ms	100%	[Original]
Teacher FP16 / BF16	Weight Half-Precision	421M	842.6 MB	-50.0%	368.0 ms	100%	[Balanced]
ONNX INT8 (Per-Channel)	Dynamic INT8 Quantization	421M	571.9 MB	-66.1%	134.7 ms	100%	[Top Recommended]
ONNX INT4 (Block-32)	4-bit Weight Quantization	421M	441.2 MB	-73.8%	680.9 ms	100%	[Slow on CPU]
ONNX INT4 (Block-64)	4-bit Weight Quantization	421M	419.6 MB	-75.1%	1047.6 ms	75%	[Degraded]
Distil-Laya 14L (FP32)	50% Depth Distillation	244M	978.0 MB	-42.0%	195.2 ms	100%	[Intermediate]
Distil-Laya 14L (INT8)	Distilled Student + INT8	244M	332.5 MB	-80.3%	78.4 ms	100%	[High Speed]
Distil-Laya 6L (FP32)	Compact Student (6 Layers)	143M	573.6 MB	-66.0%	94.0 ms	92%	[Compact]
Distil-Laya 6L (INT8)	Distilled Student + INT8	143M	195.2 MB	-88.4%	38.6 ms	92%	[Ultra Fast]
Distil-Laya 6L (INT4)	Distilled Student + INT4	143M	142.1 MB	-91.6%	112.5 ms	88%	[Ultra Small]

7. Architectural Takeaways & Deployment Recommendations

Tier 1: Maximum Accuracy (Zero Loss)	ONNX INT8 (Per-Channel) [572 MB] Shrinks storage by 66.1% (1.7 GB -> 572 MB) with 100% categorical accuracy . Leveraging Intel AVX-512 VNNI hardware units, latency improves by 2.8x (135 ms vs 382 ms). Best for general production on Intel/AMD x86 CPUs.
Tier 2: High Throughput & Balanced	Distil-Laya 14L + INT8 [332 MB] Combines 50% layer distillation with 8-bit quantization. Storage shrinks by 80.3% (1.7 GB -> 332 MB), retains 100% accuracy , and cuts latency to 78 ms (4.9x faster than teacher). Ideal for high-concurrency routing.
Tier 3: Ultra-Low Footprint & Edge	Distil-Laya 6L + INT8 / INT4 [142 - 195 MB] Reduces storage by 88.4% to 91.6% (down to 142 MB). Unlocks 38.6 ms latency (10x faster than FP32 teacher) with 92% accuracy. Perfect for edge gateways, mobile runtimes, or systems with strict memory quotas.
Avoid on x86 CPU	Teacher INT4 (Block-Wise) While reducing file size to 410-441 MB, CPU software bit-unpacking penalties cause latency to spike to 680 - 1,047 ms (5x to 8x slower than INT8), with accuracy dropping to 75% at block_size=64.